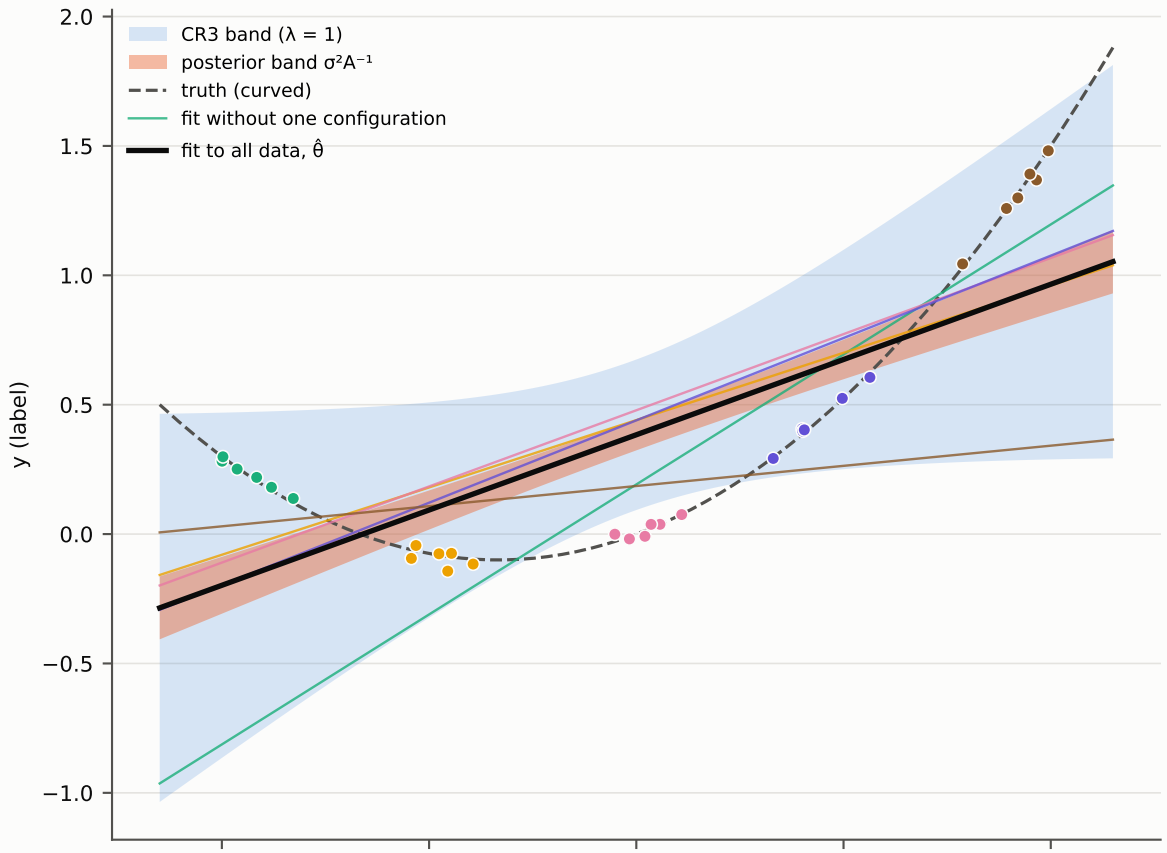
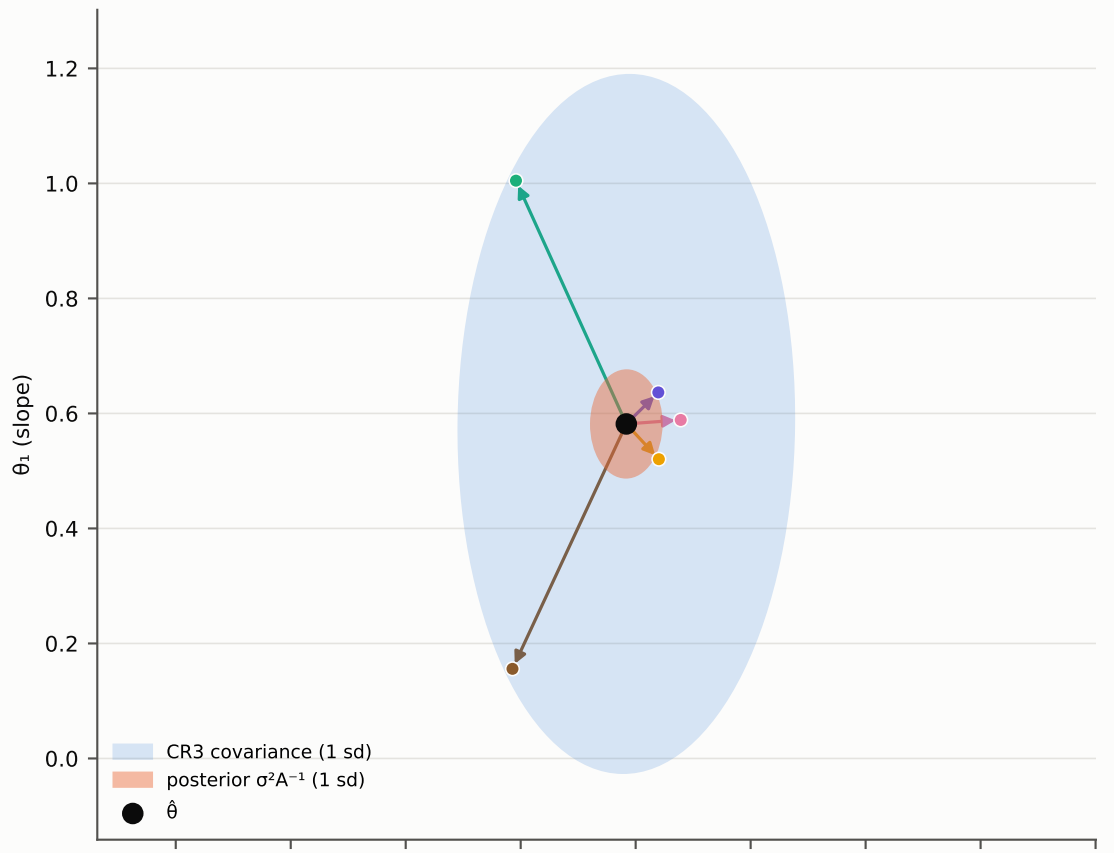
(a) A misspecified linear fit: clusters of points are configurations



(b) Parameter space: each arrow is $\hat{\theta}_{(-k)} - \hat{\theta} = -A^{-1}\tilde{g}_k$ (exact, no refit)



(c) The CR3 computation: one fit, then closed-form leave-one-configuration-out quantities

1 Fit and residuals

$$A = \Psi^T\Psi + \Lambda$$

$$\rho_k = \tilde{y}_k - \Psi_k\hat{\theta}$$

one fit to all configurations; rows Ψ_k of configuration k (energy, forces, virial) move together

2 Leverage of k

$$H_{kk} = \Psi_k A^{-1}\Psi_k^T$$

how strongly the fit is pulled towards k: eigenvalues in [0, 1); near 1 means k is nearly interpolated

3 Leave-k-out residuals

$$(I - H_{kk})^{-1}\rho_k$$

the residuals k would have if it were left out of the fit (PRESS): no refit needed

4 Pull and deletion shift

$$\tilde{g}_k = \Psi_k^T(I - H_{kk})^{-1}\rho_k$$

$$\hat{\theta} - \hat{\theta}_{(-k)} = A^{-1}\tilde{g}_k$$

the pull of k on θ as an unseen configuration would exert it; the exact shift of the fit (panel b)

5 Spread of the shifts

$$\tilde{M} = \sum_k (\tilde{g}_k - \bar{g})(\tilde{g}_k - \bar{g})^T$$

$$\Sigma_{\mathrm{CR3}} = A^{-1}\tilde{M}A^{-1} = LL^T$$

how much the fit depends on which configurations were in it: the shape, wide where the model is misspecified

6 Use, with one scale λ

$$\sigma(b) = \lambda\,\|L^T b\|$$

λ from held-out data; per atom the 3×3 force block V; QoIs by the delta method and committees in θ